TEAM ZOLLERQU

Portfolio Selection with QAOA



Which stocks to buy? One qubit per asset, sampled on the IonQ Forte 1 noise model.

Oussama Turki · Tushar Pandey

This effort is supported by Qollab & IonQ

PROBLEM

Which assets should a fund buy?

- Banks and funds build portfolios every day
- They want low risk and a good return
- Risk comes from assets that fall together

Every extra asset doubles the choices

AAPL Apple	**GLD** gold	**XOM** oil	**JPM** bank
TLT US bonds	**JNJ** health	**PG** consumer	**XLU** utilities
VNQ real estate	**FCX** copper	**MSFT** Microsoft	**SLV** silver

4,096
portfolios, 12 assets

536,870,912
portfolios, 29 assets

Data: weekly prices of 12 US stocks and funds, 2006-2014 (Kaggle Huge Stock Market Dataset, CC0)

FORMULATION

One qubit per asset, one Ising energy

Score to minimise: risk minus return

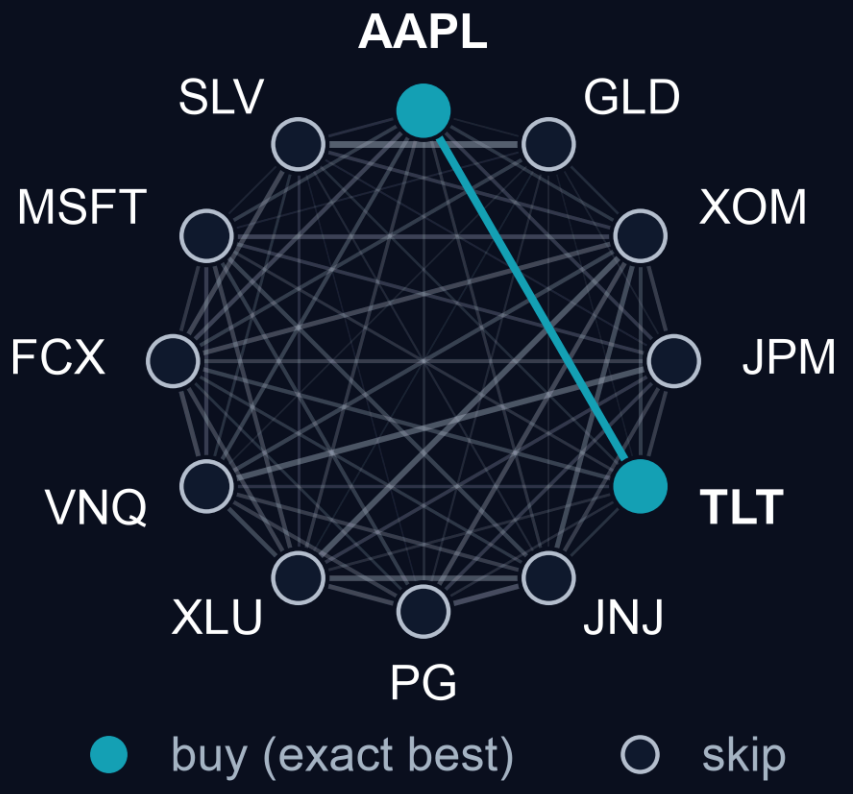
$$J(x) = \underbrace{\frac{x^{\mathrm{T}} C\, x}{n^2\, V_{\mathrm{ref}}}}_{\text{risk}} - \eta \underbrace{\frac{\mu^{\mathrm{T}} x}{n\, \sigma_{\mathrm{ref}}}}_{\text{return}}$$

The same score as an Ising energy

$$E(s) = \sum_{i<j} J_{ij}\, s_i\, s_j + \sum_i h_i\, s_i + c$$

1 qubit = 1 asset: |0> buy, |1> skip

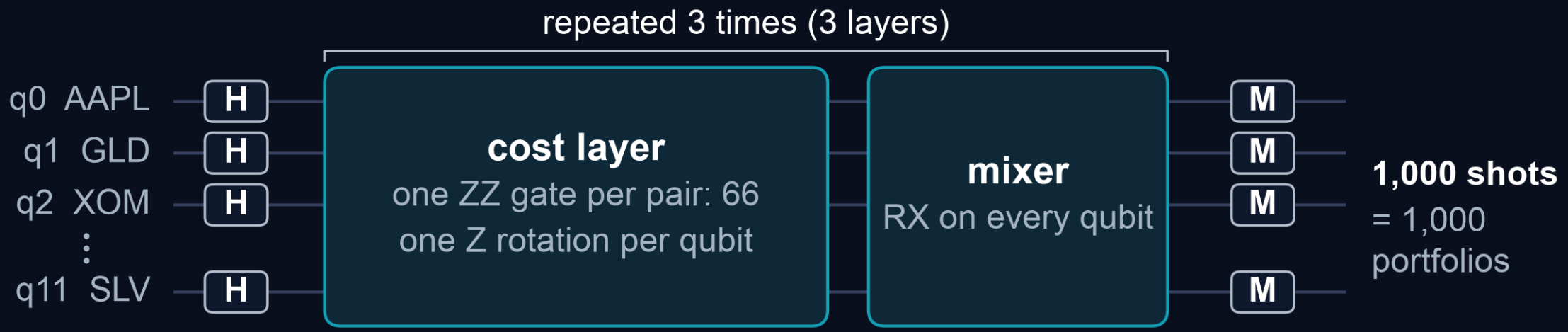
12 qubits · 66 couplings · 4,096 states



Exact best portfolio found by checking all 4,096 portfolios (classical); risk and return from 2006-2014 data

QUANTUM METHOD

QAOA circuits, trained offline, one IonQ job



7 method families

58 circuits compared, from linear-ramp and CVaR QAOA to imaginary-time evolution

Trained offline

Angles tuned on an exact classical simulator, never on the IonQ backend

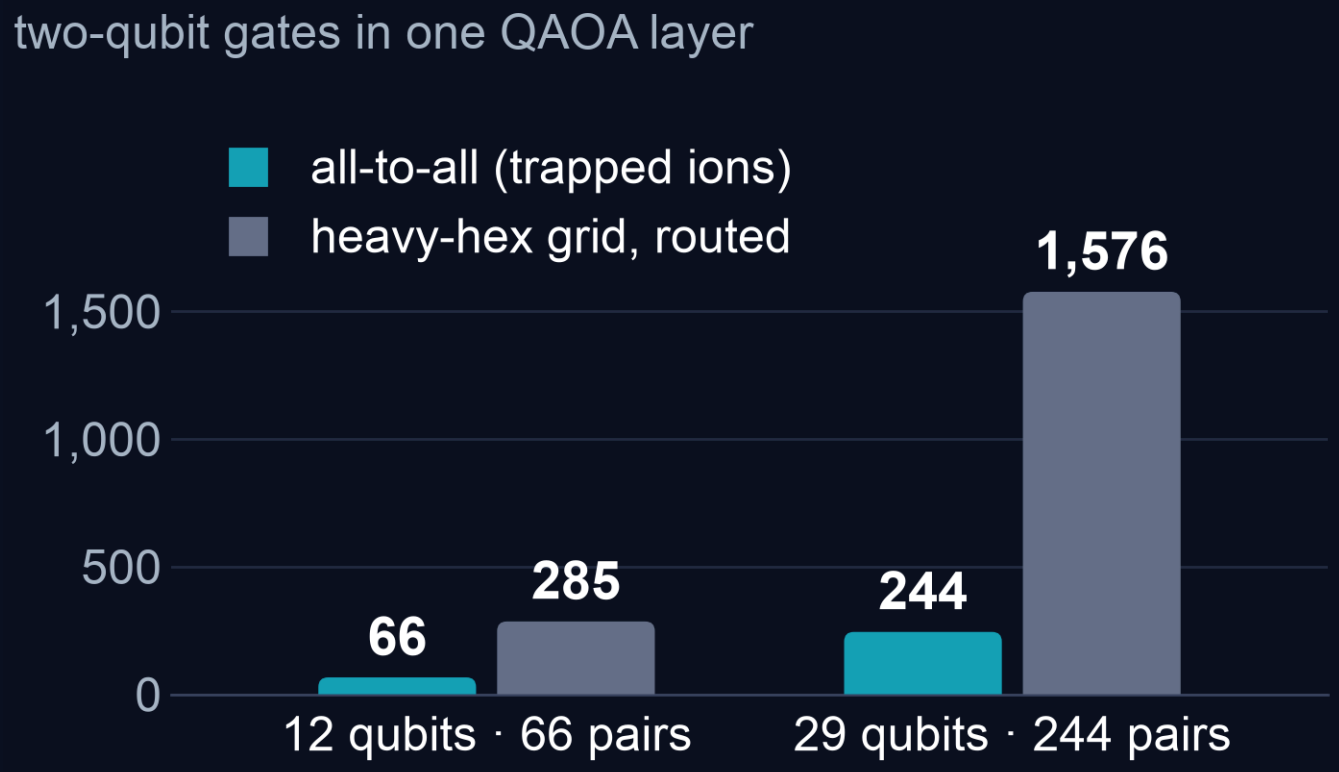
One circuit per job

CVaR-QAOA, 3 layers: 12 qubits, 198 two-qubit gates, 1,000 shots

58 circuits and a random-guess control scored by noiseless simulation (server), 12 qubits

WHY IONQ

All-to-all links: 4.3× fewer two-qubit gates



4.3×

fewer two-qubit gates
at 12 qubits (66 vs 285)

6.5×

fewer at 29 qubits (244 vs 1,576)

66 pairs = 66 gates, no SWAPs

Gate counts from transpiling only (Qiskit 2.5.2, level 1); heavy-hex = median of 5 routing seeds on a generic heavy-hex map. No circuit was run on a heavy-hex device.

MAIN RESULT

IonQ noise model: 29× random guessing

IonQ Forte 1 noise model

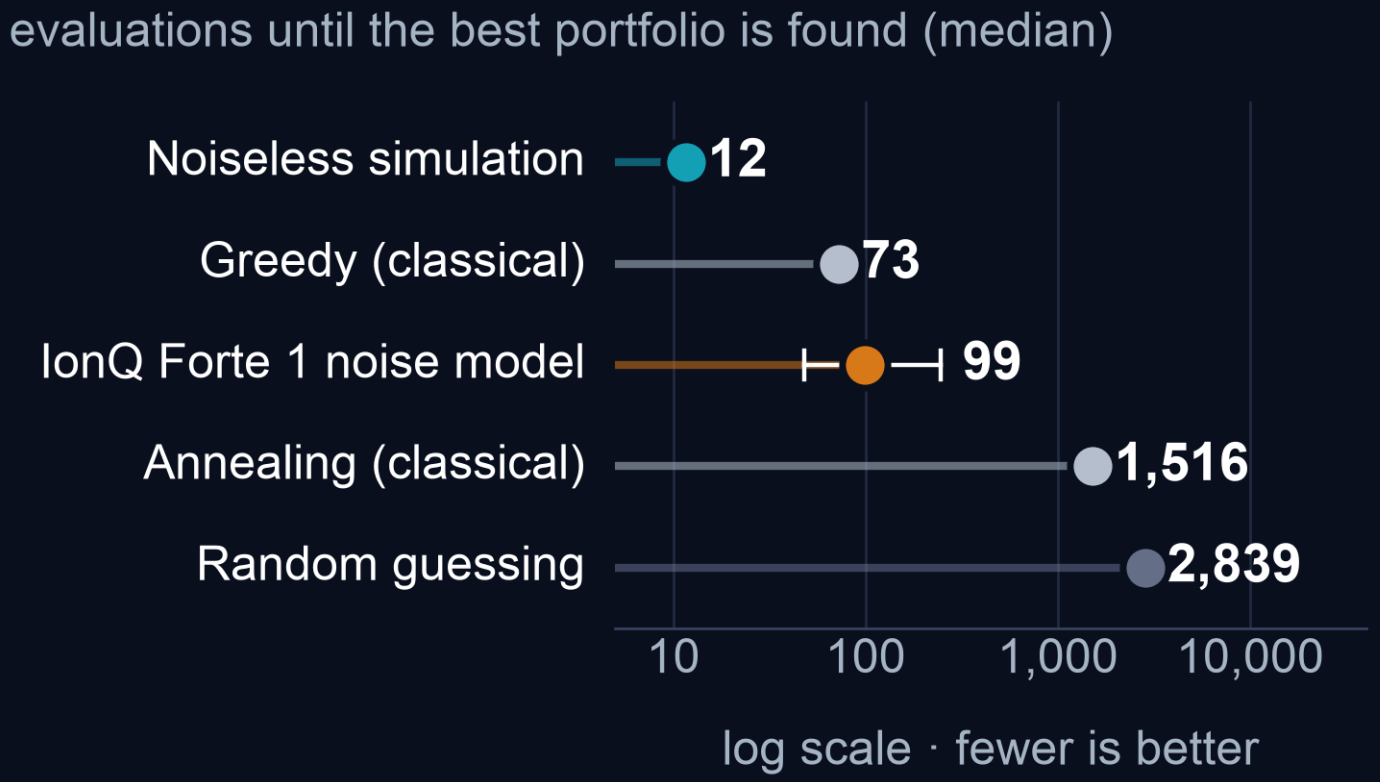
7 / 1,000

shots are the exact best of 4,096 portfolios

29× random guessing

58 / 1,000 noiseless

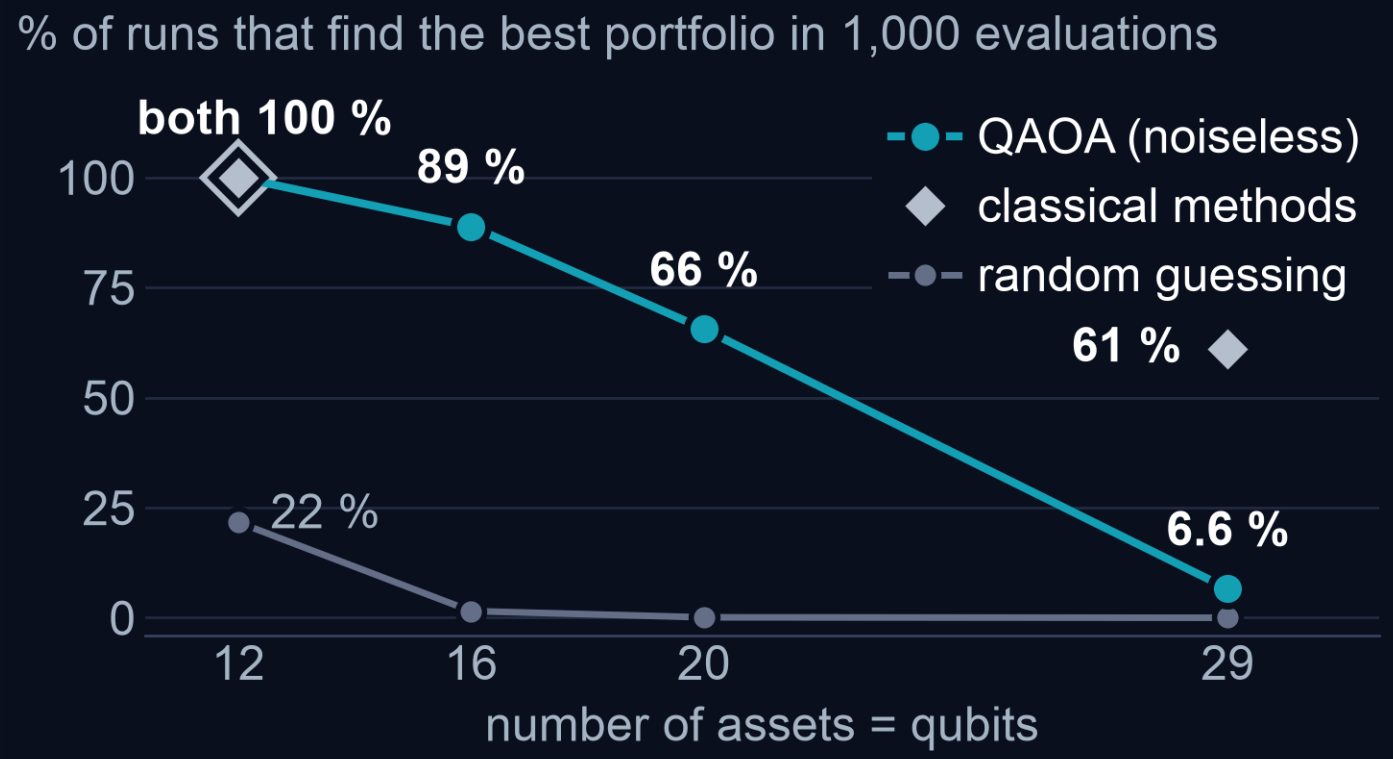
95 % interval: 3-14 per 1,000



IonQ Forte 1 noise model (Qollab): 1 job, 1,000 shots, 11 Oct 2026 · noiseless simulation
Classical: median of 50 runs. Sampler medians come from the hit rate per shot. Training not counted.

SCALING AND HONEST COMPARISON

Up to 29 qubits, classical heuristics win



Read with care

- Training came first: 4,860 simulated circuits
- A shot is not the same work as a classical evaluation
- 12 assets: greedy needs 73 evaluations, our noisy run 99
- 29 qubits: noiseless simulation only, angles reused from 12

12-20 qubits: server simulation, 10 Oct 2026 · 29 qubits: server simulation of all 536,870,912 states, 11 Oct 2026 · classical: 200 runs · the same 1,000 evaluations for every method

DEMO, LIMITS, NEXT STEPS

Try it on Qollab: one click, one IonQ job

[pending] screenshot

Qollab → Run → Select QPU →
IonQ Forte 1 (29q) noise model

Qollab project: [pending]
GitHub: [pending]
Video: [pending]

This effort is supported by Qollab & IonQ

Honest limits

- A noisy simulator: one job of 1,000 shots
- Noise cuts the hits from 58 to 7 per 1,000
- Classical heuristics match or beat it

Next steps

- Run the other 4 selected circuits
- Shorter circuits: 66 instead of 198 gates
- 29 assets on the IonQ Forte 1 noise model

APPENDIX

One engine, two problems

	Stocks: portfolio selection	Mesh: split for 2 devices
Qubits	12	12
Circuit sent to IonQ	CVaR-QAOA, 198 gates	varQITE ZY, 66 gates
Best answer, noise model	7 / 1,000 shots	278 / 1,000 shots
Random guessing	0.24 / 1,000	0.49 / 1,000
Gain over random guessing	29×	569×
Noiseless simulation	58 / 1,000	750 / 1,000

This effort is supported by Qollab & IonQ

IonQ Forte 1 noise model (Qollab): one 1,000-shot job per project, 11 Oct 2026 · noiseless: simulation
Stocks: results/ionq_B_cvar_p3_*.json · Mesh: mesh/qollab/results_ionq/*.json on the mesh branch